

EXP:1 SETTING UP THE ENVIRONMENT AND PREPROCESSING THE DATA

AIM

To preprocess the Titanic dataset by handling missing values, encoding categorical variables, performing feature scaling, and splitting the dataset into training and testing sets for machine learning applications.

PROCEDURE

1. Import the required Python libraries such as Pandas, NumPy, Seaborn, and Scikit-learn.
2. Load the Titanic dataset using the Seaborn library.
3. Display the first few records and explore the dataset using information, statistical summary, and missing value analysis.
4. Handle missing values by filling the **Age** column with the median value and the **Embark Town** column with the mode value.
5. Remove the **Deck** column due to a large number of missing values.
6. Encode the categorical columns **Sex** and **Embark Town** using Label Encoding.
7. Remove unnecessary columns that are not required for model training.
8. Standardize the numerical features **Age** and **Fare** using StandardScaler.
9. Split the dataset into training and testing sets using the train-test split method.
10. Display the shape of the training and testing datasets.

CODE:

```
import pandas as pd
import numpy as np
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler

df = sns.load_dataset("titanic")

print(df.head())
```

```
print(df.info())
print(df.describe())
print(df.isnull().sum())

df["age"] = df["age"].fillna(df["age"].median())
df["embark_town"] = df["embark_town"].fillna(df["embark_town"].mode()[0])

df.drop(columns=["deck"], inplace=True)

le = LabelEncoder()

df["sex"] = le.fit_transform(df["sex"])
df["embark_town"] = le.fit_transform(df["embark_town"])

df.drop(columns=["embarked", "class", "who", "alive", "adult_male", "alone"],
inplace=True)

scaler = StandardScaler()

df[["age", "fare"]] = scaler.fit_transform(df[["age", "fare"]])

X = df.drop("survived", axis=1)
y = df["survived"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
```

)

```
print("Training Data Shape:", X_train.shape)
```

```
print("Testing Data Shape:", X_test.shape)
```

```
print(X_train.head())
```

OUTPUT:

```
---
0      survived  pclass    sex  age  sibsp  parch    fare  embarked  class  \
1      0         3      male  22.0    1     0    7.2500    S  Third
2      1         1      female  38.0    1     0   71.2833    C  First
3      1         3      female  26.0    0     0    7.2500    S  Third
4      1         1      female  35.0    1     0   53.1000    S  First
5      0         3      male   35.0    0     0    8.0500    S  Third

0      who  adult_male  deck  embark_town  alive  alone
1  woman      False    C  Cherbourg    yes  False
2  woman      False  NaN  Southampton    yes   True
3  woman      False    C  Southampton    yes  False
4    man      True  NaN  Southampton    no   True
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   column              Non-Null Count  Dtype
---  -
 0   survived            891 non-null     int64
 1   pclass              891 non-null     int64
 2   sex                 891 non-null     object
 3   age                 714 non-null     float64
 4   sibsp               891 non-null     int64
 5   parch               891 non-null     int64
 6   fare                891 non-null     float64
 7   embarked            889 non-null     object
 8   class               891 non-null     category
 9   who                 891 non-null     object
10  adult_male          891 non-null     bool
11  deck                289 non-null     category
12  embark_town          889 non-null     object
13  alive                891 non-null     object
14  alone                891 non-null     bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 88.7+ KB

None
survived    pclass    age    sibsp    parch    fare
count  891.000000  891.000000  714.000000  891.000000  891.000000  891.000000
mean    0.383838    2.308642    29.699118    0.523008    0.381594    32.204208
std     0.486592    0.436071    14.526497    1.182743    0.886657    49.693429
min     0.000000    1.000000    0.420000    0.000000    0.000000    0.000000
25%     0.000000    2.000000    20.125000    0.000000    0.000000    7.910400
50%     0.000000    3.000000    28.000000    0.000000    0.000000    14.454200
75%     1.000000    3.000000    38.000000    1.000000    0.000000    31.000000
max     1.000000    3.000000    60.000000    8.000000    6.000000    512.329200

survived    0
pclass      0
sex         177
age         0
sibsp       0
parch       0
fare        0
embarked    2
-----
2
```

```
parcn      0
fare       0
embarked    2
class      0
who         0
adult_male  0
deck       688
embark_town 2
alive       0
alone       0
dtype: int64
Training Data Shape: (712, 7)
Testing Data Shape: (179, 7)

pclass  sex  age  sibsp  parch  fare  embark_town
331     1   1  1.240235    0    0 -0.074583    2
733     2   1 -0.488887    0    0 -0.386671    2
382     3   1  0.282762    0    0 -0.488854    2
784     3   1 -0.258337    1    0 -0.490280    2
813     3   0 -1.795334    4    2 -0.018709    2
```

RESULT

The Titanic dataset was successfully preprocessed by handling missing values, encoding categorical variables, performing feature scaling, and splitting the dataset into training and testing sets. The processed dataset is now ready for machine learning model development.